



FitPredict: Activity Prediction and Motivation System

A Proactive Approach to Calorie Prediction using LSTMs

IIT Ropar - Capstone Project

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The Problem: From Reactive Data to Proactive Coaching

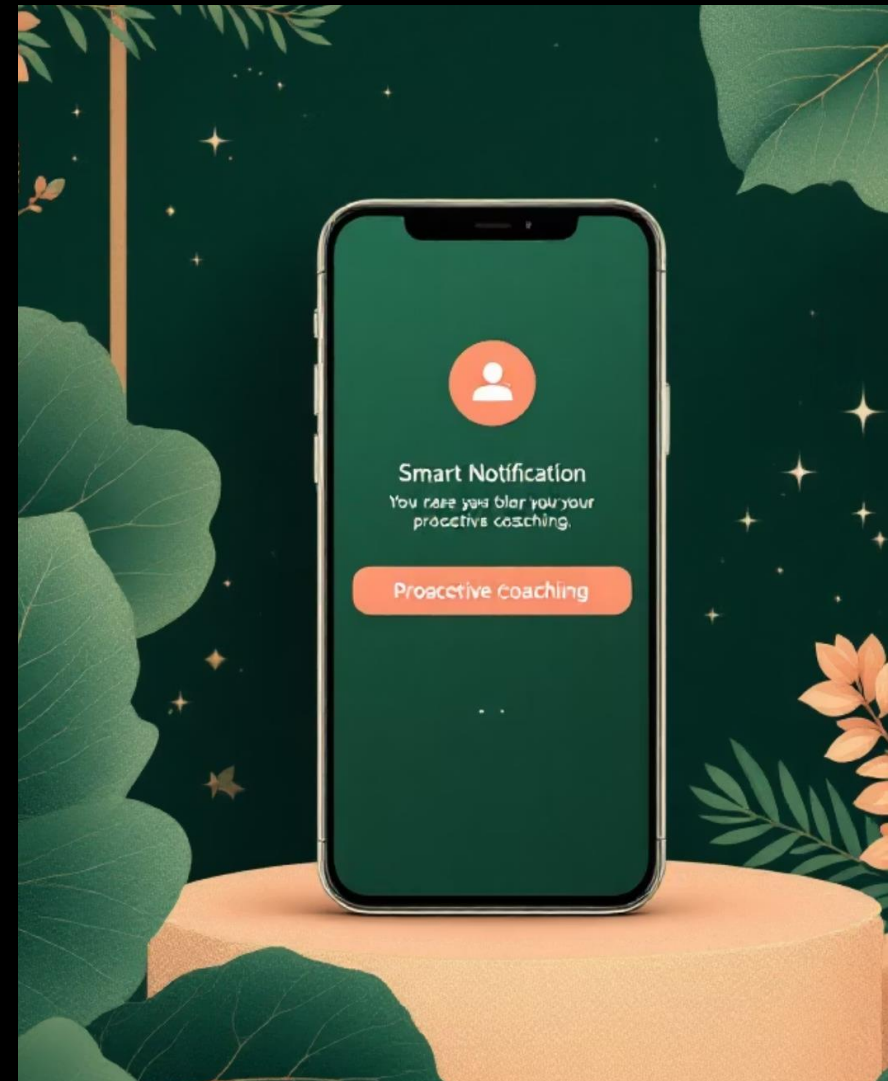
Current Reality

- Fitness trackers are reactive - show what you've already done
- Users check stats at day's end when it's too late
- Creates "all or nothing" cycle that demotivates

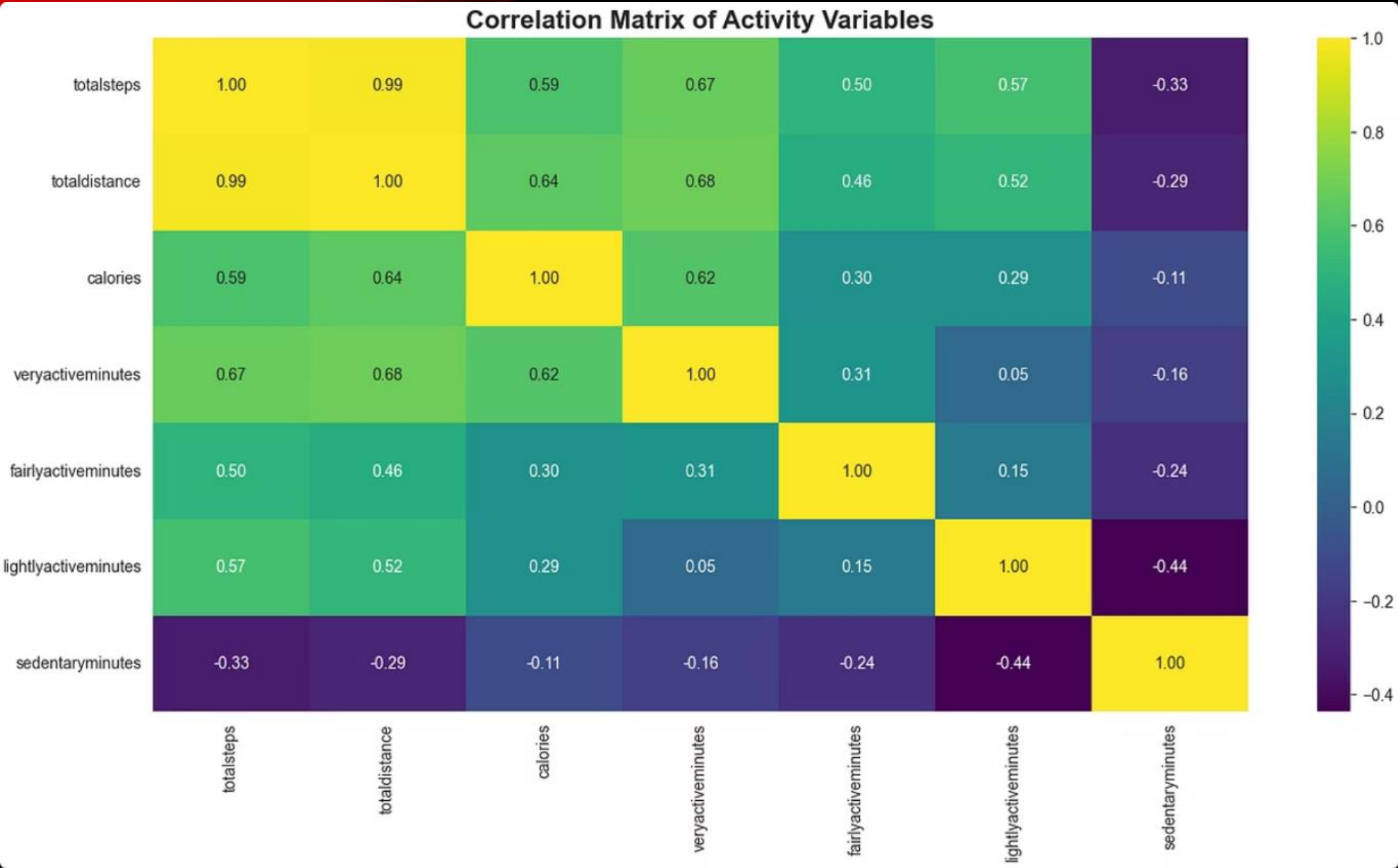


Our Solution

- Proactively forecast end-of-day results
- Provide early YES/NO verdict on daily goals
- Enable timely adjustments for success



Dataset & Key Insights



Data Source

FitBit Fitness Tracker Data

33 Users, 1 Month

Source: Kaggle

Granularity

Daily, Hourly & Minute-level data

Calories, Steps, Intensities, Sleep, Heart Rate

Key Discovery

The most direct way to increase overall activity is by reducing sedentary time.

Distinct user personas identified

Analysis revealed the need for personalized prediction models based on distinct behavioral patterns.

Methodology: Five-Phase Pipeline



Data Preprocessing

Cleaned, transformed and merged 1.5 Million+ data points



Feature Engineering

Created hourly features from minute-sleep (sleep quality) and heart rate data (Stress level)



Model Training

Built LSTM model with early stopping optimization



Daily Forecasting

Iterative prediction script for daily total projection



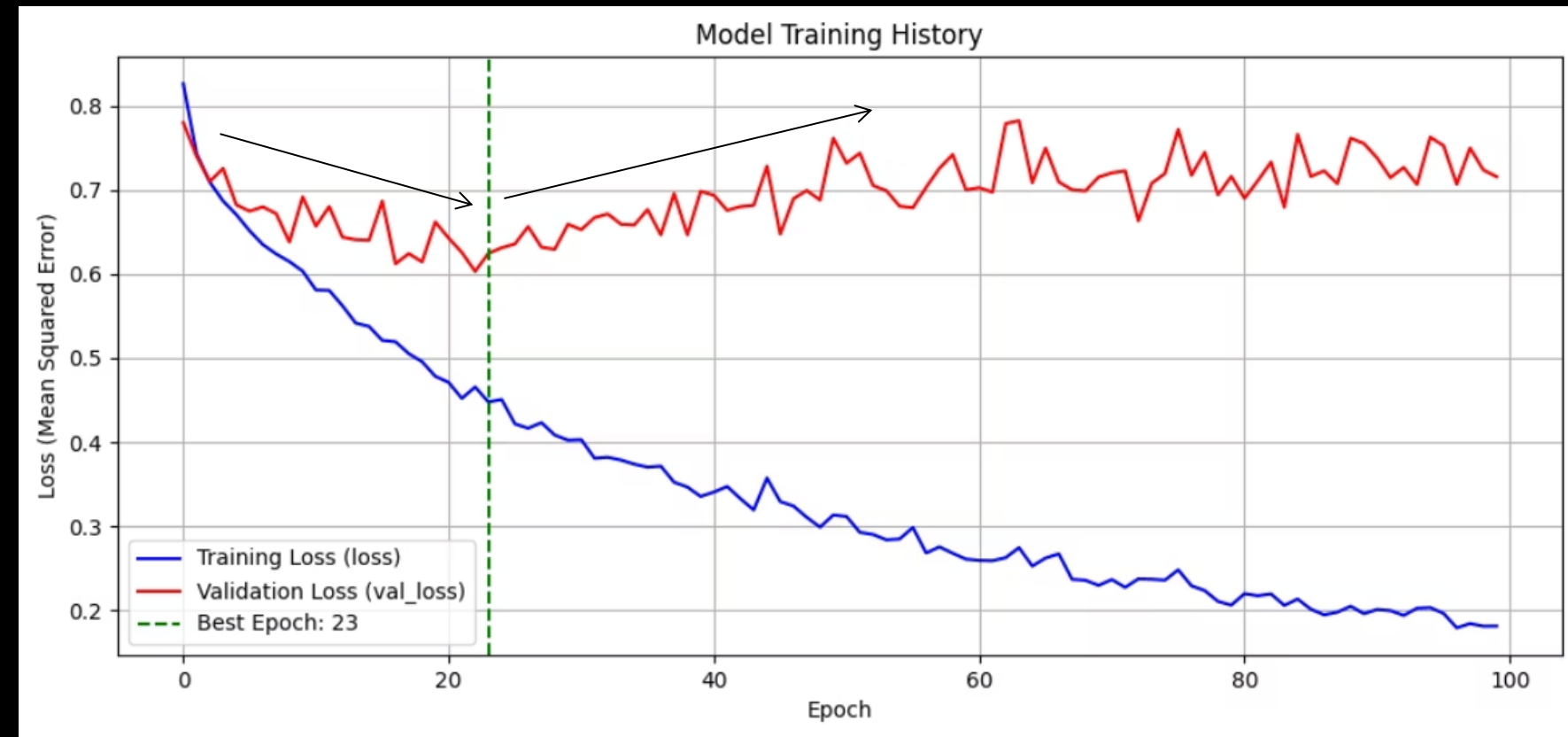
Evaluation

Rigorous testing on unseen test dataset

The Model: LSTM for Time-Series Forecasting

Why LSTM?

- Recurrent Neural Network ideal for sequential data
- Memory capability learns from past 24 hours
- Perfect for predicting next hour's calories



Architecture

01

Input

Define model input
(24×N Features)

02

LSTM Layer 1

Processes data sequences (64)

03

Dropout

Randomly deactivates 20% neurons

04

LSTM Layer 2

Processes the last sequence (32)

05

Dense Layer

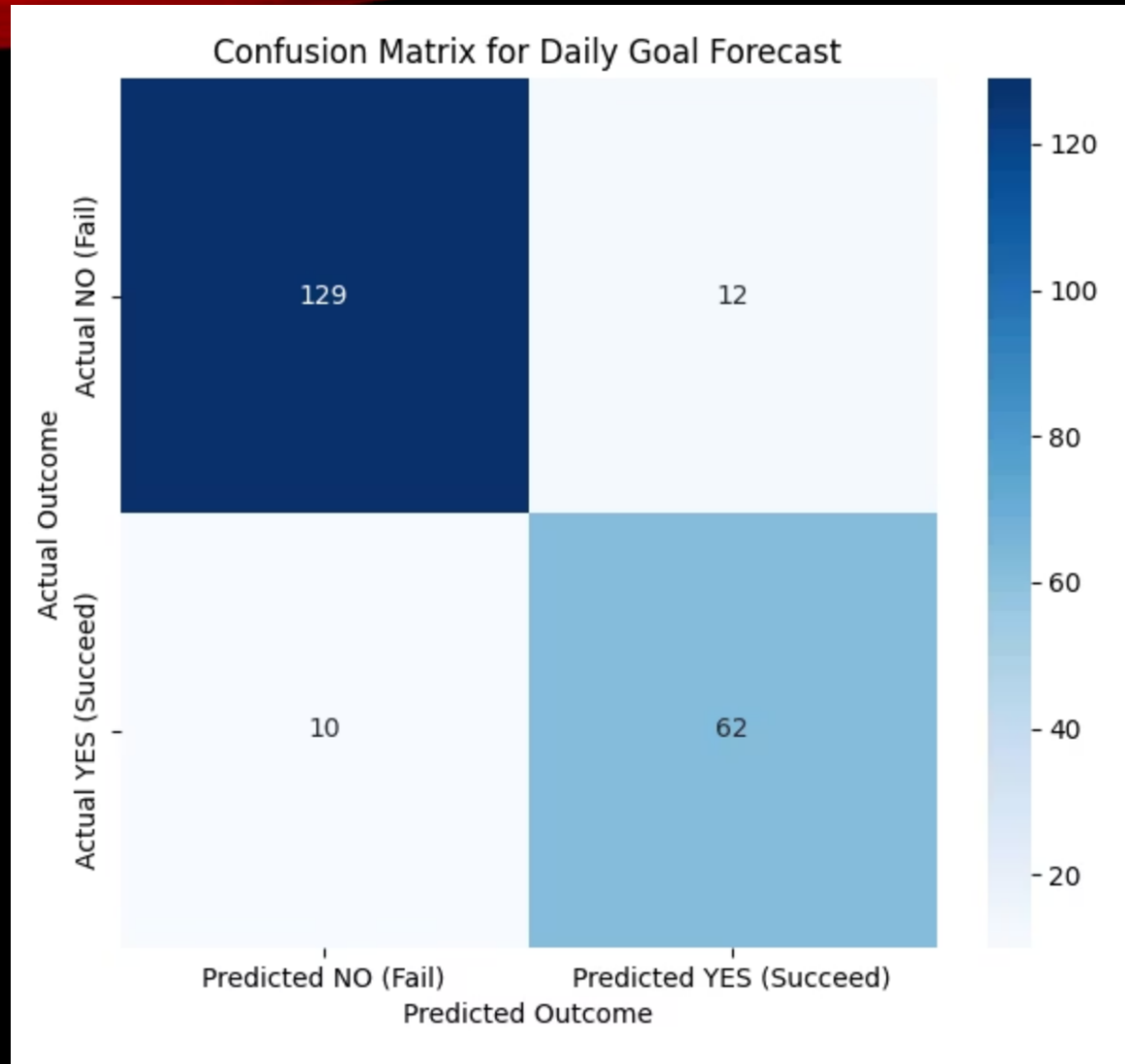
Processes final output (16, "Relu")

06

Output

Generates final prediction
(1 Calorie Value)

Results: Classification Performance



89.67%

Accuracy

Correct YES/NO verdict 9 out of 10 times

83.78%

Precision

When predicting YES, right 84% of time

86.11%

Recall

Correctly identifies 86% of successful days

84.93%

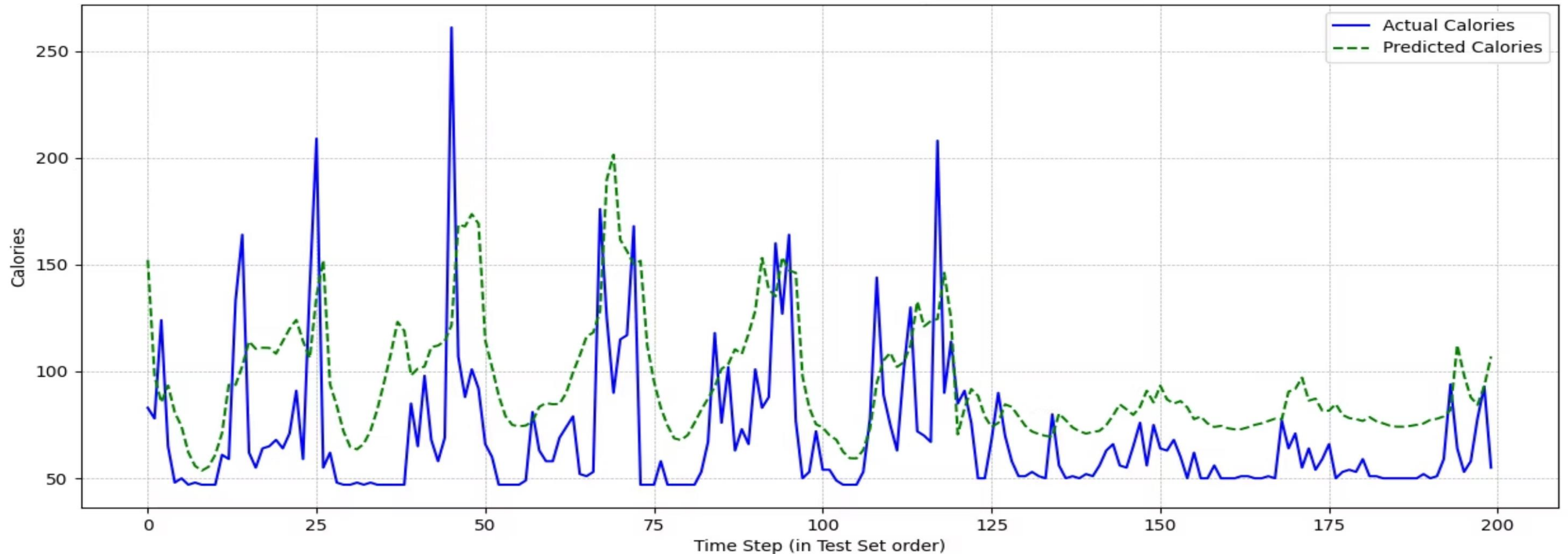
F1-Score

Strong balanced performance indicator



The model provides a highly reliable and trustworthy daily goal achievement forecast.

Model Performance: Actual vs. Predicted (First 200 Test Points)



Regression Analysis: Calorie Projection Accuracy

Key Metrics

- **MAE:** 334.82 calories
- **RMSE:** 513.03 calories

Key Insight

Model excels at predicting daily direction but struggles with peak event magnitude. Higher RMSE indicates larger errors on spontaneous high-intensity days.

Conclusion: Directional accuracy explains why classification performs significantly better than raw regression.

Key Takeaways



Proactive Success

Built reliable LSTM forecaster achieving ~90% accuracy in daily goal prediction



EDA Drives Strategy

In-depth user analysis revealed behavioral personas, justifying personalized approach



Direction Over Magnitude

Model strength lies in classifying day type (active vs inactive) for motivational coaching



Real-World Ready

Saved model and scripts form robust foundation for user-facing application

Future Enhancements



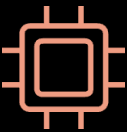
Generative AI Coaching

Integrate forecast with GPT-4 for dynamic, personalized motivational messages instead of static verdicts.



User-Facing Application

Deploy model via Flask/FastAPI backend powering mobile app with real-time coaching dashboard.



Advanced Models

Enhance accuracy with Heart Rate Variability features and explore Transformer architectures.





Thank You

Questions?

Ready to discuss the future of proactive fitness coaching